Individual Assignment Cover Sheet

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I. Introduction

The present document provides an assessment of the potential consequences for BP (British Petroleum P.L.C.) on embracing the United Nations Sustainable Development Goals (SDGs). Specifically, the opportunities and risks goals address in this work will be centred around how the UN Targets are related to Clean Energy, Innovation, Sustainable Consumption and Climate Action. The key decision for BP to make is whether they would benefit from expanding their deployed EV charging infrastructure and embrace SDGs based on identified opportunities and risks.

Energy Provisioning for Transport has been selected as main industry for this assessment given its CO2 global impact and BP is chosen given its worldwide relevance and intention to improve its overall environmental reputation. According to a study on greenhouse emission worldwide (Statista, 2024, p.25), the global C2 Emissions in 2023 were caused by Transportation 21.11% and Power Industry by 38.24%. Based on these 2 metrics the energy provision for transport plays a significant role on global C02 emissions.

II. Motivation

The motivation for this assessment comes from 2 main drivers for BP: Public Relations and long-term commercial strategy. First, given the prevalence of the ecological impact 2010 Gulf of Mexico spill, BP as a global organisation has committed more than 69 Billion USD to compensate for the economic impact and legal settlements (BP America, 2024). More recently, BP has committed towards their given the significant discussion of climate change impact and acknowledging new opportunities for BP to improve their global image, they have publicised their sustainability commitments as shown in their Net Zero Ambition Report (BP P.L.C., 2024, p2).

The second driver comes from a commercial strategy standpoint. The development of sustainable Electric Vehicles (EVs) charging infrastructure is critical to place the company in a leading position for this growing market, which in turn is linked to a significant reduction of carbon impact. According to a worldclass sustainability report (Statista, 2023, p. 12) the usage of EVs has displaced 13.1 billion litres of gasoline with a significant carbon emissions reduction. BP through a global engagement on EV infrastructure development would embrace four critical UN SDGs.

III. UN Framework in Sustainable Development

The United Nations originally defined the Sustainable Development Goals in 2015 derived from the Millenium Development Goals (MDGs). The SDGs as per the UN published references (United Nations, 2024) remained as a more specific set of meaningful targets for all its signing countries to achieve true and sustainable development across all nations, with emphasis on environmental protection and responsible economic development. This set of principles can represent a unique

opportunity for BP to consolidate their commitments on sustainable business practices, together with solid corporate accountability and transparency.

Given the priorities mentioned in the Sustainability Report (BP P.L.C. 2023) the organisation is reinventing itself to embrace new technologies and NetZero frameworks. As acknowledged in the document, there are different angles to evaluate the best sustainable approach to enable this meaningful transition. By embracing four specific UN SDGs, discussed in the next section, BP will structure their sustainability and corporate actions on NetZero and green development.

IV. Assessment on each SDG

The SDGs found to be more relevant for the Net Zero Goals and Sustainable Strategy of BP are introduced in this section based on the UN SDG main site (United Nations, 2024). Each SDG is then contrasted with the areas of opportunity and risk that each will represent to BP.

SDG 7 - Affordable and Clean Energy. Under this goal the targets are defined towards renewable energy mix (target 7.2), energy efficiency (target 7.3) and global cooperation on research and development of cleaner sources of energy (target 7.a).

Considerations – In order to develop affordable and clean energy solutions for EV charging infrastructure, BP could expand its investments in offshore wind energy, in Northern Europe where the company already possesses considerable infrastructure and expertise according to their reports (BP P.L.C., 2023). The company might also explore innovative energy storage solutions, such as integrating hydrogen production facilities with its wind farms to create a more reliable renewable energy system. Additionally, they could design their EV charging network by integrating solar panels and battery storage at charging stations. This would let renewable energy to power vehicles while providing better experience for customers.

SDG 9 - Industry Innovation and Infrastructure. This goal specifies the targets around infrastructure as follows: developing sustainable and reliable infrastructure (target 9.1) and using clean and environmentally sound technologies (target 9.4).

Considerations – For further innovation in the EV charging domain, BP could implement smart-grid technology to optimise charging times based on grid carbon intensity and introduce mobile charging vans for emergency top-ups in remote areas. To improve the customer experience and maximise station efficiency, the company could develop an AI-powered app that predicts queue times and suggests optimal charging routes, and then partner with retailers to provide quality amenities during the charging wait time.

SDG 12 - Ensure sustainable consumption and production patterns. This goal primarily is focused on guiding multinational organisations to adopt sustainable practices and integrate sustainability data into their reporting processes (target. 12.6).

Considerations – The company can enhance sustainable consumption and production patterns through several targeted initiatives. For instance, they could significantly increase their investments in renewable energy sources as suggested in the previous paragraphs. And gradually reducing their traditional oil and gas operations. This transition should be working together with more efficient production processes that reduces waste and environmental impact across their existing operations, including the adoption of carbon capture and storage technologies. BP could improve its approach to solar energy by developing cutting-edge solar technologies integrated with smart grid systems and use the continuously growing windfarm infrastructure deployed across Europe and USA.

SDG 13 - Climate Action. This goal emphasises the climate changes measures to be implemented strategically and with thorough planning (target 13.2).

Considerations – BP can take climate action efforts by strategically expanding its EV charging network, particularly focusing on rapid-charging hubs at existing petrol stations and working with local authorities to install chargers in urban areas with limited off-street parking. The company should prioritise powering these charging points with renewable energy as previously exposed in the other SDGs using renewable energy sources. Meanwhile, it is possible to implement a dynamic pricing model that encourages off-peak charging to reduce grid strain. Additionally, integrating digital solutions that allow drivers to easily locate available chargers and pay seamlessly would help accelerate EV adoption, consequently furthering BP's contribution to emissions reduction.

V. Analysis of Sustainability Challenges and Opportunities

Government Guidelines

Opportunities – Both in North America and Europe there are ambitious guidelines and policies from their respective governments to maximise the deployment of EV infrastructure. For instance, in UK, the NetZero Mandate (UK Government, 2024) specifies that by 2030 80% of all new vehicles need to be running on non-fossil fuels. Similar initiatives are defined in USA and mainland Europe (Statista, 2024).

Risks – Political environments can lead to a diminishing interest to provide sufficient support on the EV journey adoption, hence compromising the value of the returns on the investments to be done by BP on charging infrastructure. For instance, are strong changes on the UK government support on EV infrastructure according to the ruling party (Godar, 2024). Nevertheless, the overall commitment to remove fossil fuel dependency is a long-term objective as reported by the global emissions report (Statista, 2024).

Growing Market of EVs

As reported by Statista (2023, p9) the global market of EVs is growing significantly, leading to a consistently growing demand for charging infrastructure. It is expected to quadruple in size between 2021 and 2025.

Opportunities – With the previously mentioned growth, there is a solid justification on investing on further infrastructure, with a significant increase on revenue through the respective energy provisioning.

Risks – According to Statista (2023, p9) there is still range anxiety, upfront costs and lack of charging infrastructure that can prevent the EV charging points market to grow as expected. BP, together with its competitors, EV manufacturers and governments need to collaborate to mitigate these risks and manage driver expectations accordingly.

Renewable Energy Supply

It is important to note that renewable energy sources need to be provision proportionally to the growth of EV infrastructure planning. It is claimed that EVs generate 66% less of carbon emissions compared to their fossil fuel counter parts (Local Government, 2024).

Opportunities – There is good availability of renewable energies that BP can already connect to their supply grid as per the current number of chargers. This in turn can leverage additional technologies as explored in the SDG 7 subsection, for enable solar panels to be collocated with the charge points and reduce losses on electricity transmission through the grid.

Risks – There is a clear challenge on whether renewable energy supply would scale by 20x as it is potentially the expected increase once most of the EV charging points will be deployed in the upcoming future. Absence of sufficient renewable energy supply would lead to the need for additional dependency on coal, gas, nuclear energy and waste incineration to compensate. Consequently, the emissions of the EVs could significantly grow or become close to the emissions of the fossil fuel powered vehicles.

Competition

Currently, Shell aims for 90% of drivers to be within 10 minutes' drive of a Shell rapid charger by 2030 (Shell U.K. Limited, 2024). BP needs to react and plan with respect to their main competitor accordingly.

Opportunities – Competitors as Shell are moving quickly towards deploying a large infrastructure for the provision of energy and deployment of charging points across US, Europe and UK at an aggressive pace. BP can aim to catch up at a similar pace considering similar goals of been very close to most of the drivers in these developed countries.

Risks – As infrastructure will need to grow by at least one order of magnitude, there are significant challenges on setting up the equipment in the right locations and ensure that there is not an oversupply in particular areas and undersupply in other areas. Furthermore, talent could be constrained due to the required amount of training and compensation required for a solid maintenance force serving this infrastructure.

VI. Connection to other sustainability frameworks

Natura is a successful global cosmetics company headquartered in Brazil. Although their industry differs from BP, still their corporate governance can bring significant light on sustainability and accountability practices to BP. According to public article on Natura (Centre for Climate Engagement Climate Governance, 2023), they have introduced the Integral Profit and Loss Methodology that accounts on collecting data across P&L reporting focused on environmental and social impact. Their methodology can easily be implemented by BP to quantifying the environmental effect of the SDGs and EV energy provisioning previously discussed.

Another relevant methodology, on validating the full NetZero accountability to be reported by BP is the concept of Regeneration. As described in the Stanford Social Innovation Review (Marquis, 2024), regeneration consists of letting natural cycles to let the planet replenish the resources we have consumed. This is illustrated with a renewable energy organisation located in Vermont, with unique market and regenerative mechanisms to generate carbon offsets and decrease financial difficulties. In the context of BP, regeneration together with Integral P&L methodologies, can lead to ensure BP can certify their energy sources to be fully renewable and regenerative. These methodologies in turn lead to a fully integrated NetZero approach on EV energy distribution.

One final remark, in terms of potential global sustainability recognition, BP can aim to set the example for their peers and adjacent industry players. There is a 2025 Global 100 is a ranking on sustainability (Corporate Knights, 2024). Their energy methodology quantifies metrics such as the energy use vs the renewable energy generated by a company. Aiming to be ranked top by the Corporate Knights, would let BP to fully go ahead with the NetZero strategy for their EV charging infrastructure, and embrace again the SDGs.

VII. Strategic Recommendations

Based on the previous analysis, BP should scale up their EV charging infrastructure. Despite the challenges and risks mentioned before, the investment presents a big strategic opportunity. The global shift towards transport electrification, assisted by governmental policies and growing EVs demand, suggests that companies such as BP must adapt their business models to remain relevant in a rapidly evolving energy

landscape. BP's existing supply chain of renewable energy resources is a key advantage to serve the expanding EV market.

The company's established brand recognition, aided by its deep expertise in energy distribution, places it in a good position to capture a significant share of the EV charging market. The integration of EV charging infrastructure would also complement BP's broader strategy of diversifying into renewable energy, helping to future-proof its business against declining fossil fuels demand (Statista, 2024).

VIII. Conclusion

Despite challenges such as grid capacity constraints and scaling renewable energy supply chain, early investment in EV infrastructure could lead to substantial long-term benefits. By establishing a larger presence in the EV charging space, BP can build better global reputation, embrace UN SDGs, and potentially influence industry standards. To do so, BP needs to: 1) quantify their EV charging energy supply with Integral P&L; 2) validate their energy sources Regeneration; and 3) aim for the highest levels of sustainability recognition as illustrated with the Corporate Knights ranking. Furthermore, the company can leverage its expertise in global energy distribution and mitigate the risks derived from competition, change of government EV incentives, changes on EV demand and renewable energies supply. This will lead to a more sustainable and resilient business model for the time to come.

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